



Incident report analysis

Instructions

As you continue through this course, you may use this template to record your findings after completing an activity or to take notes on what you've learned about a specific tool or concept. You can also use this chart as a way to practice applying the NIST framework to different situations you encounter.

Scenario

Review the scenario below. Then complete the step-by-step instructions.

You are a cybersecurity analyst working for a multimedia company that offers web design services, graphic design, and social media marketing solutions to small businesses. Your organization recently experienced a DDoS attack, which compromised the internal network for two hours until it was resolved.

During the attack, your organization's network services suddenly stopped responding due to an incoming flood of ICMP packets. Normal internal network traffic could not access any network resources. The incident management team responded by blocking incoming ICMP packets, stopping all non-critical network services offline, and restoring critical network services.

The company's cybersecurity team then investigated the security event. They found that a malicious actor had sent a flood of ICMP pings into the company's network through an unconfigured firewall. This vulnerability allowed the malicious attacker to overwhelm the company's network through a distributed denial of service (DDoS) attack.

To address this security event, the network security team implemented:

- A new firewall rule to limit the rate of incoming ICMP packets
- Source IP address verification on the firewall to check for spoofed IP addresses on incoming ICMP packets
- Network monitoring software to detect abnormal traffic patterns
- An IDS/IPS system to filter out some ICMP traffic based on suspicious characteristics

As a cybersecurity analyst, you are tasked with using this security event to create a plan to improve your company's network security, following the National Institute of Standards and Technology (NIST) Cybersecurity Framework (CSF). You will use the CSF to help you navigate through the different steps of analyzing this cybersecurity event and integrate your analysis into a general security strategy. We have broken the analysis into different parts in the template below. You can explore them here:

- **Identify** security risks through regular audits of internal networks, systems, devices, and access privileges to identify potential gaps in security.
- **Protect** internal assets through the implementation of policies, procedures, training and tools that help mitigate cybersecurity threats.
- **Detect** potential security incidents and improve monitoring capabilities to increase the speed and efficiency of detections.
- **Respond** to contain, neutralize, and analyze security incidents; implement improvements to the security process.

Recover affected systems to normal operation and restore systems data and/or assets that have been affected by an incident.

Summary	The organization experienced a DDoS attack which compromised the internal network for 2 hours. This caused the organization's network services to stop responding because of a flood of ICMP packets as well as normal internal traffic could not access any network resources. The incident management team responded by blocking the incoming ICMP packets and found that the attacker was able to send a flood of ICMP pings to the company's network through an unconfigured firewall. This allowed the malicious attacker to
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	overwhelm the company's network with a DDoS attack. To address the security event, the network security team implemented a new rule rate limiting the amount of ICMP pings on the firewall, source IP addresses verification, network monitoring software to detect unusual traffic patterns and a IDS/IPS system to filter out some ICMP packets.
Identify	The team found that a DDoS attack had occurred which overwhelmed the company's network services and was due to an unconfigured firewall. The malicious actor took advantage of this and flooded the company's network with a bunch of ICMP pings and overwhelmed the company.
Protect	The team implemented a new firewall rule that limits the number of ICMP packets that come through the firewall. They also added source IP address verification to prevent spoofing of the IP addresses on the incoming ICMP packets. Furthermore, there is network monitoring software implemented to detect unusual traffic patterns as well as an IDS/IPS system in place to filter our ICMP traffic based on suspicious characteristics.
Detect	To detect future abnormalities,there is an IDS/IPS in place that will flag suspicious incoming traffic and a network monitoring software to detect abnormal traffic patterns.
Respond	The team blocked the incoming ICMP packets which stopped all non-critical network services and restored critical network services. They investigated the issue and found a malicious actor had sent a flood of ICMP packets through an unconfigured firewall. For the future, the team will isolate the affected systems and attempt to restore critical systems and then the team will analyze network logs to check for suspicious activities and report all incidents to upper management and legal authorities if needed.
Recover	The team was able to store critical services after blocking the ICMP packets and implemented newer rules to monitor for future changes. In the future, ICMP flood attacks will be blocked by the firewall and all non-critical network

	services should be stopped or reduced. Critical services should be restored first and once the flood of ICMP packets have timed out, all non-critical services can be brought back online.
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Reflections/Notes:
